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Practice : 1

In [6]:
```
from sklearn.datasets import load_breast_cancer

data = load_breast_cancer(as_frame=True)
df = data.frame
print(df.shape)
df.head()
```

(569, 31)

Out[6]:

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	s
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	

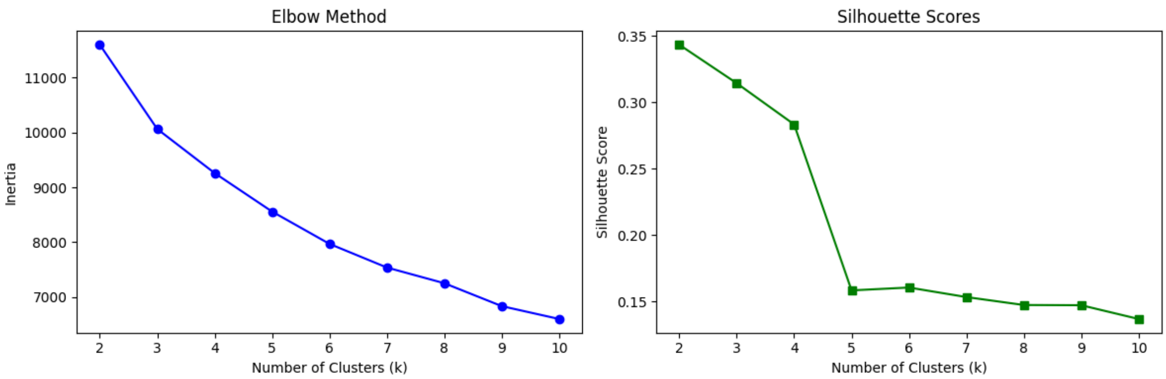
5 rows × 31 columns

In [7]:
```
from sklearn.preprocessing import StandardScaler

X = df.drop(columns=['target']).dropna()

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print(f"Shape after preprocessing: {X_scaled.shape}")
```

Shape after preprocessing: (569, 30)

In [8]:
```
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

k_range = range(2, 11)
inertias = []
sil_scores = []

for k in k_range:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = km.fit_predict(X_scaled)
    inertias.append(km.inertia_)
    sil_scores.append(silhouette_score(X_scaled, labels))

# Elbow Plot
plt.figure(figsize=(12, 4))
```

```python
plt.subplot(1, 2, 1)
plt.plot(k_range, inertias, 'bo-')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Inertia')
plt.title('Elbow Method')

# Silhouette Plot
plt.subplot(1, 2, 2)
plt.plot(k_range, sil_scores, 'gs-')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Silhouette Score')
plt.title('Silhouette Scores')

plt.tight_layout()
plt.show()

best_k = list(k_range)[sil_scores.index(max(sil_scores))]
print(f"Best k by Silhouette: {best_k} (score: {max(sil_scores):.4f})")
```



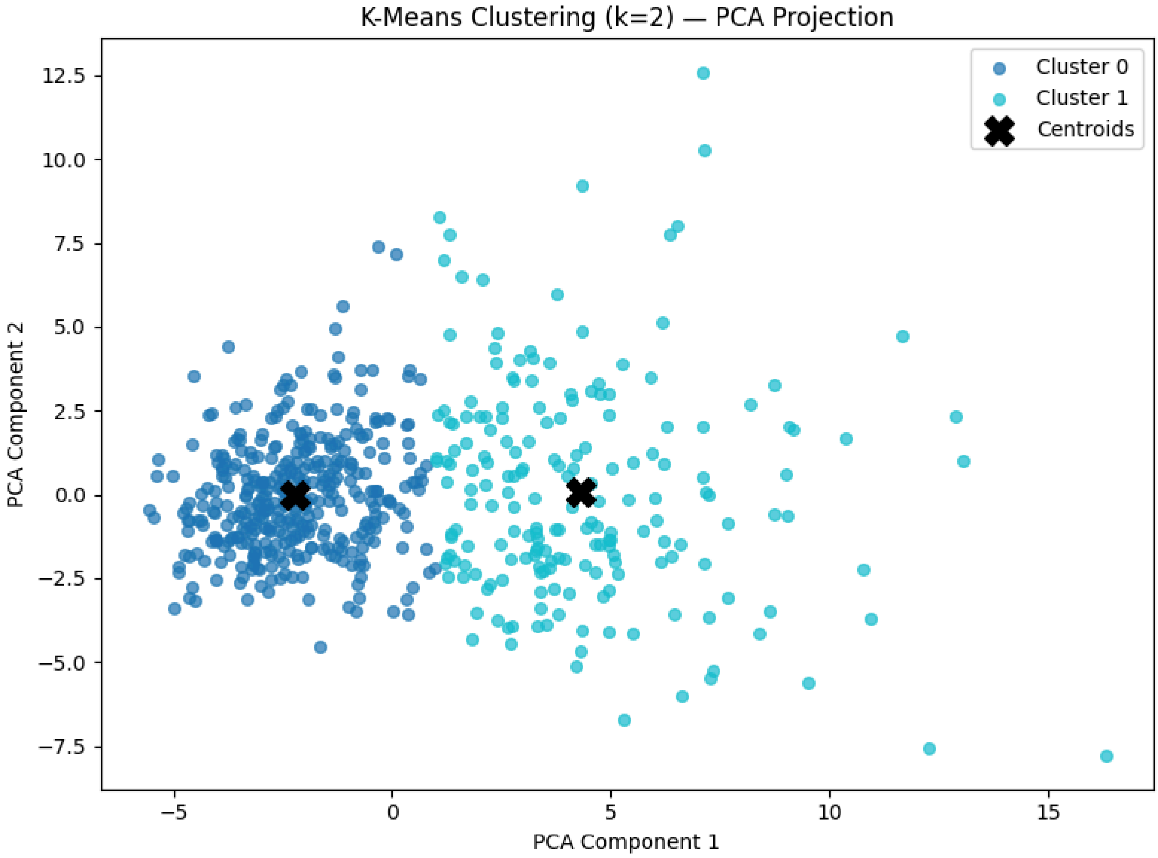

```
Best k by Silhouette: 2 (score: 0.3434)
```

In [9]:
```python
kmeans = KMeans(n_clusters=best_k, random_state=42, n_init=10)
cluster_labels = kmeans.fit_predict(X_scaled)

print(f"KMeans trained with k={best_k}")
print(f"Inertia: {kmeans.inertia_:.2f}")
```

```
KMeans trained with k=2
Inertia: 11595.53
```

In [10]:
```python
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
import numpy as np

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# Plot clusters
plt.figure(figsize=(8, 6))
colors = plt.cm.tab10(np.linspace(0, 1, best_k))

for i in range(best_k):
    pts = X_pca[cluster_labels == i]
    plt.scatter(pts[:, 0], pts[:, 1], s=30, color=colors[i], label=f'Cluster {i}

# Plot centroids projected to PCA space
centroids_pca = pca.transform(kmeans.cluster_centers_)
plt.scatter(centroids_pca[:, 0], centroids_pca[:, 1], s=200, c='black', marker='
```

```
plt.title(f'K-Means Clustering (k={best_k}) — PCA Projection')
plt.xlabel('PCA Component 1')
plt.ylabel('PCA Component 2')
plt.legend()
plt.tight_layout()
plt.show()
```



In [11]:
```
import pandas as pd

results = pd.DataFrame({'k': list(k_range), 'Inertia': inertias, 'Silhouette Scc
print(results.to_string(index=False))

drops = [inertias[i] - inertias[i+1] for i in range(len(inertias)-1)]
elbow_k = list(k_range)[drops.index(max(drops)) + 1]

print(f"\nElbow Method suggests k = {elbow_k}")
print(f"Silhouette Method suggests k = {best_k}")
print(f"\nBest k selected: {best_k}")
print(f"Final Silhouette Score: {silhouette_score(X_scaled, cluster_labels):.4f}
print("\nInterpretation: A silhouette score closer to 1.0 indicates well-separat
```

k	Inertia	Silhouette Score
2	11595.526607	0.343382
3	10061.797818	0.314384
4	9258.989105	0.283305
5	8558.660667	0.158210
6	7970.263835	0.160367
7	7540.318697	0.153186
8	7254.326193	0.147195
9	6837.628936	0.147040
10	6603.404402	0.136656

Elbow Method suggests k = 3
Silhouette Method suggests k = 2

Best k selected: 2
Final Silhouette Score: 0.3434

Interpretation: A silhouette score closer to 1.0 indicates well-separated, compac
t clusters.